Azure Databricks Mastering Guide

Module 1: Architecture Setup & Enterprise Security Blueprint (Pages 1–20)

Author: Principal Cloud Data Architect

Target Audience: Data Engineers, Solutions Architects, Security Admins

Version: 2026.1 Enterprise Release

Scope: Detailed analysis of Control/Data Plane, VNet Injection, and SCC Topology

1. Control Plane & Data Plane Separation Matrix

Azure Databricks runs on a **split-plane architecture** designed to guarantee clear operational boundaries between the management stack and processing nodes. Understanding where your code executes versus where your metadata resides is essential for passing rigid enterprise security reviews.

Architectural Dimension	Control Plane (Managed by Databricks)	Data Plane (Managed in Customer Azure Subscription)
Hosted Location	Secure Databricks-owned Azure subscription.	Customer's own Azure Resource Group (Managed Resource Group).
Core Components	Notebook Web UI, Cluster Manager, Jobs API, Back-end Metastore, Identity management infrastructure.	Spark Compute Clusters, DLT Pipelines, Serverless Endpoints, Local SSDs, Mounted storage layers.
Data Exposure	Stores metadata, query definitions, telemetry, and access lists. *Never stores raw data.*	Processes and stores actual transactional records, logs, and business assets natively.
Security Responsibility	SOC 2 Type II compliance handled directly by Databricks/Microsoft.	Configured by customer via Network Security Groups (NSGs) and Firewalls.

Security Isolation Boundaries

- **Cross-Plane Encryption:** All RPC communication passing between the Data Plane and Control Plane runs via TLS 1.2/1.3 tunnels anchored with dynamic symmetric keys.
- **No Inbound Public IPs:** The Control Plane coordinates work without mapping public IPs to backend processing workers when Secure Cluster Connectivity is active.

2. VNet Injection & Secure Cluster Connectivity (SCC)

Why Enterprise Architectures Mandate VNet Injection:

1. **Private Storage Endpoints:** Connect securely to Azure Storage Accounts via Azure Private Endpoints without allowing traffic out to the open internet.

2. **On-Premises Connectivity:** Allows Spark worker nodes to connect back to on-premises data sources via Azure ExpressRoute or Site-to-Site VPN tunnels.

3. **Custom Firewalls:** Routes all outbound traffic through an Azure Firewall or centralized Network Virtual Appliance (NVA) for inspection and data exfiltration filtering.

The Subnet Allocation Architecture

Subnet Profile	Delegation Class	Minimum CIDR	Functional Objective

3. Workspace Infrastructure Network Troubleshooting Matrix

Error Code / Symptom	Root Cause	Resolution Strategy
Status: Metastore Connection Timeout		
Status: Cluster Creation Failed		
Status: AuthorizationDenied		

4. Production Infrastructure Automation Deployment Blueprint

```
{
  "$schema": "https://schema.management.azure.com/schemas/2019-04-01/deploymentTemplate.jso
n#",
  "contentVersion": "1.0.0.0",
  "parameters": {
    "workspaceName": { "type": "string", "defaultValue": "prod-databricks-wsp" },
    "pricingTier": { "type": "string", "defaultValue": "premium" },
    "customVnetId": { "type": "string" },
    "customPublicSubnetName": { "type": "string", "defaultValue": "db-public-sn" },
    "customPrivateSubnetName": { "type": "string", "defaultValue": "db-private-sn" }
  },
  "resources": [
    {
      "type": "Microsoft.Databricks/workspaces",
      "apiVersion": "2024-05-01",
      "name": "[parameters('workspaceName')]",
      "location": "[resourceGroup().location]",
      "sku": { "name": "[parameters('pricingTier')]" },
      "properties": {
        "managedResourceGroupId": "[concat(subscription().id, '/resourceGroups/', parameter
s('workspaceName'), '-managed-rg')]",
        "parameters": {
          "customVirtualNetworkId": { "value": "[parameters('customVnetId')]" },
          "customPublicSubnetName": { "value": "[parameters('customPublicSubnetName')]" },
          "customPrivateSubnetName": { "value": "[parameters('customPrivateSubnetName')]" }
,
          "enableNoPublicIp": { "value": true }
        }
      }
    }
  ]
}
```

Configuration Checklist for Production Go-Live:

- **enableNoPublicIp = true:** This flag activates Secure Cluster Connectivity, removing public IP anchors from individual compute VMs.
- **Premium Sku Mandatory:** Required to enable Unity Catalog integrations, detailed cluster ACL policies, and single sign-on attribute routing mapping configurations.